

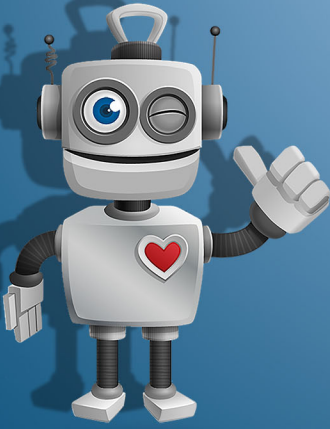
تقدم لكم اللجنة الاكاديمية في فريق

Mechatronics spectrum

امتحان سابق لمادة

الاحكامات المنطقية المبرمجة

الفصل الأول ٢٠١٦



فيرست



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Student Name:

Q1) Circle the most accurate answer for the following statement (15 marks)

- 1- A relay output PLC has an analog to digital convertor with 8 bit for a temperature sensor input channel. The temperature range is 100°C to 400°C . The resolution of the temperature reading is:
☒ a- 1.17°C b- 1.56°C c- 0.39°C d- None
- 2- A free contact switch (regular switch) is to be wired to a PLC. It should be wired as:
a- Sinking b- Sourcing ☒ c- Sinking or sourcing d- transistor output.
- 3- The most common type of PLC outputs is
☒ a- Relay output, as the existence of relays as outputs makes it easier to connect with external devices.
b- Transistor output, as the transistors are very fast c- Triac output, as it can supply ac loads.
d- None.
- 4- One of the following is not true:
a- Most PLCs are used with small applications
☒ b- Most PLCs I/O are digital.
c- Most of PLC challenges are solved within 20 rung ladder.
d- None.
- 5- One of the followings is not true
☒ a- An output contact can be used many times in the ladder program.
b- Each rung in the ladder can contain only one output.
c- Contact cannot be directly placed on the right of output. d- None.
- 6- The device that combines the functionality of a PC and reliability of a PLC is:
a- Touch Screen ☒ b- Programmable Automation Controller c- Single-Board Computer d- None.
- 7- Chose a correct statements:
a- PLCs are similar to conventional computers in terms of hardware technology.
☒ b- PLCs were invented to replace Electromechanical control parts with solid state ones.
c- A&B d- None.

8- A Triac output PLC is:

- A. Non-polarized and typically can switch either AC or DC. ✗
- B. Non-polarized and typically can switch only DC. ✗
- C. Polarized and typically can switch either AC or DC. ✗
- D. None.**

9- One of the following is not a disadvantage of Relay-based control

- a- Too much work required in connecting wires ✗
- b- Difficulty with changes or replacements ✗
- c- Needs power supply with specific power levels..**
- d- None

10- With rotating magnetic hard drives and non-industrially hardened components, such as PLC power supplies, PCs were more prone than PLCs to failure.

- a- True b- False

1	2	3	4	5	6	7	8	9	10
A	C	A	B	A	B	B	D	C	A

Q2) Draw the output (Q0.0 & Q0.1) of the following Ladder diagram. (3 marks)

